D3V3NDRA CLASSES

INTERNATIONAL MATHEMATICS OLYMPIAD (IMO) • TRAINING SERIES

SPECIAL EDITION • LOGICAL REASONING MASTER EXAMINATION

Class 7 (Grade VII) • Specialized Cognitive Reasoning Syllabus • 50 MCQs • 100 Marks

Candidate Name: ..... **Roll No:** ..... **School / Center:** ..... **Date:** .....

Time: 60 Mins | **Max Marks:** 100 | **Total Questions:** 50 MCQs | **Marking:** +2 / -0 (No Negative) | **Curriculum:** 8 Cognitive Domains • 16 Diagrammatic MCQs (Mirror & Water Images)

SECTION 1: MIRROR IMAGES & LATERAL REFLECTIONS (DIAGRAMMATIC)

Q1. Choose the correct mirror image in the vertical mirror line shown.

Given	A	B	C	D
				

(A) Option A **(B) Option B** **(C) Option C** **(D) Option D**

Q2. Choose the correct mirror image in the vertical mirror line shown.

Given	A	B	C	D
				

(A) Option A **(B) Option B** **(C) Option C** **(D) Option D**

Q3. Choose the correct mirror image in the vertical mirror line shown.

Given	A	B	C	D
				

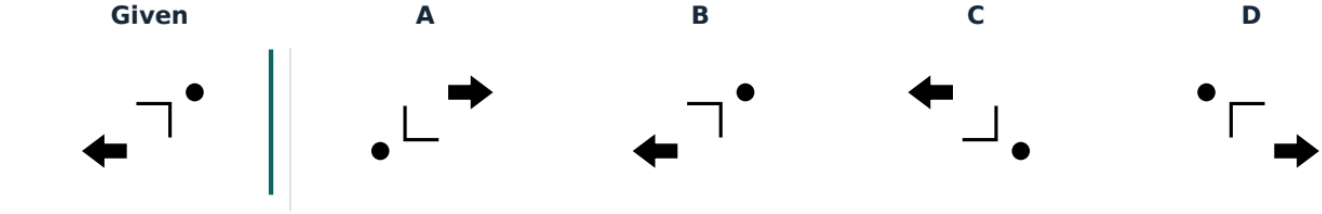
(A) Option A **(B) Option B** **(C) Option C** **(D) Option D**

Q4. Choose the correct mirror image in the vertical mirror line shown.

Given	A	B	C	D
				

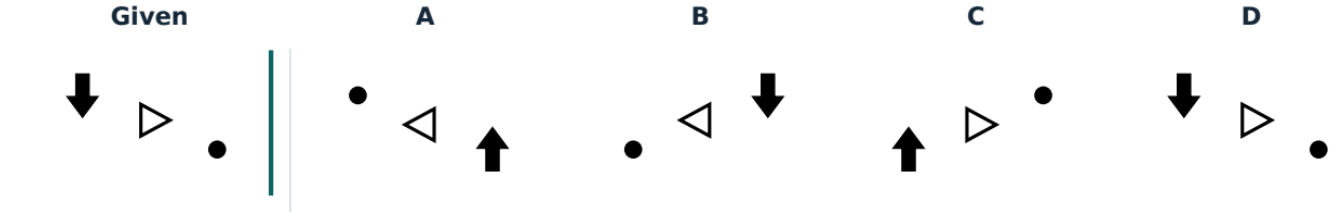
(A) Option A **(B) Option B** **(C) Option C** **(D) Option D**

Q5. Choose the correct mirror image in the vertical mirror line shown.



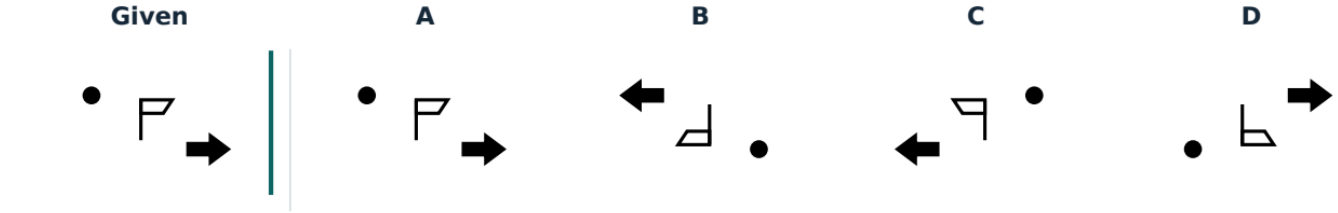
(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q6. Choose the correct mirror image in the vertical mirror line shown.



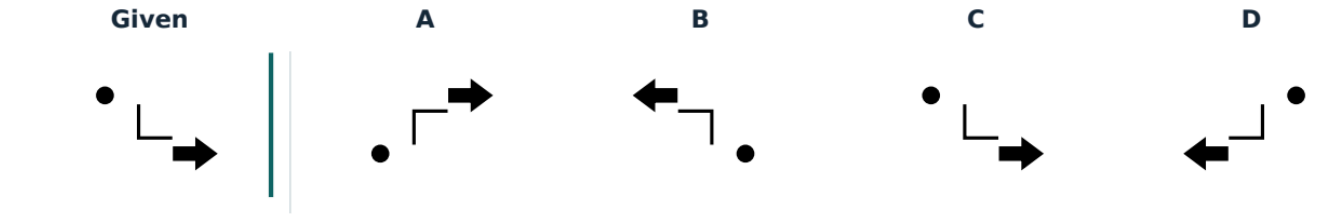
(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q7. Choose the correct mirror image in the vertical mirror line shown.



(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q8. Choose the correct mirror image in the vertical mirror line shown.



(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

SECTION 2: WATER IMAGES & VERTICAL REFLECTIONS (DIAGRAMMATIC)

Q9. Choose the correct water image in the horizontal water line shown.

Given



A



B



C



D



(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q10. Choose the correct water image in the horizontal water line shown.

Given



A



B



C



D



(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q11. Choose the correct water image in the horizontal water line shown.

Given



A



B



C



D



(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q12. Choose the correct water image in the horizontal water line shown.

Given



A



B



C



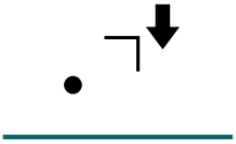
D



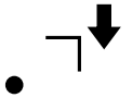
(A) Option A **(B)** Option B **(C)** Option C **(D)** Option D

Q13. Choose the correct water image in the horizontal water line shown.

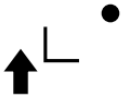
Given



A



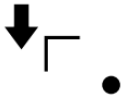
B



C



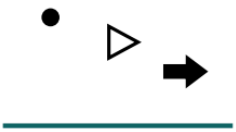
D



(A) Option A (B) Option B (C) Option C (D) Option D

Q14. Choose the correct water image in the horizontal water line shown.

Given



A



B



C



D



(A) Option A (B) Option B (C) Option C (D) Option D

Q15. Choose the correct water image in the horizontal water line shown.

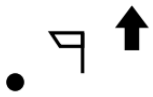
Given



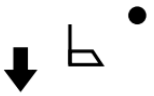
A



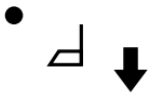
B



C



D



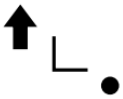
(A) Option A (B) Option B (C) Option C (D) Option D

Q16. Choose the correct water image in the horizontal water line shown.

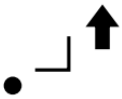
Given



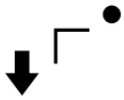
A



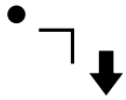
B



C



D



(A) Option A (B) Option B (C) Option C (D) Option D

SECTION 3: NUMBER & ALPHABET SERIES COMPLETION

Q17. Find the next number: 6, 11, 16, 21, 26, ?

(A) 32 **(B)** 31
(C) 36 **(D)** 30

Q18. Find the next number: 1, 4, 9, 16, 25, ?

(A) 49 **(B)** 35
(C) 30 **(D)** 36

Q19. Find the next number: 3, 7, 13, 21, 31, ?

(A) 42 **(B)** 41
(C) 45 **(D)** 43

Q20. Find the next number: 50, 47, 43, 38, 32, ?

(A) 26 **(B)** 25
(C) 27 **(D)** 24

Q21. Find the next number: 1/2, 1, 3/2, 2, 5/2, ?

(A) 4 **(B)** 7/2
(C) 3 **(D)** 11/4

Q22. Find the next number: 5, 10, 8, 16, 14, 28, ?

(A) 24 **(B)** 56
(C) 30 **(D)** 26

SECTION 4: CODING & DECODING LOGIC

Q23. Every letter is shifted forward by one place in the alphabet. If CAT is coded as DBU, how is DOG coded?

(A) EPH **(B)** FQI
(C) EOG **(D)** CNE

Q24. Every letter is shifted forward by two places. If BALL is coded as DCNN, how is GAME coded?

(A) HBNF **(B)** ICOG
(C) ICOH **(D)** EZKC

Q25. Use A=1, B=2, ..., Z=26. Which word is represented by 3-1-20?

(A) CAR **(B)** CAT
(C) CUT **(D)** BAT

Q26. A word is first reversed, then every letter is moved forward one place. For example, CAT becomes UBD. How is DOG coded?

(A) GOD **(B)** HPE
(C) EPH **(D)** IQF

Q27. Letters in positions 1 and 3 move forward two places; letters in positions 2 and 4 move backward one place. If FROG becomes HQQF, how is LION coded?

(A) NHRM **(B)** MHPM
(C) NHQM **(D)** NKQP

Q28. A word is coded by the sum of its alphabet positions (A=1, ..., Z=26). What is the code for HOME?

(A) 39 **(B)** 42
(C) 40 **(D)** 41

SECTION 5: BLOOD RELATIONS & FAMILY DEDUCTIONS

Q29. Asha is the mother of Rohan. Rohan is the father of Meera. How is Asha related to Meera?

(A) Mother **(B)** Sister
(C) Grandmother **(D)** Aunt

Q30. Rita and Sita are sisters. Rita is the mother of Kabir, and Sita is the mother of Anu. How is Kabir related to Anu?

(A) Nephew **(B)** Cousin
(C) Uncle **(D)** Brother

Q31. Neel is the brother of Kavya. Kavya is the mother of her daughter Isha. How is Isha related to Neel?

(A) Niece **(B)** Aunt
(C) Sister **(D)** Daughter

Q32. Pointing to a woman, Sameer says, "Her father is my mother's father." The woman is not Sameer's mother. How is the woman related to Sameer?

(A) Aunt **(B)** Grandmother
(C) Daughter **(D)** Sister

Q33. Pooja and Seema are sisters. Seema is the mother of her son Anil. How is Anil related to Pooja?

(A) Nephew **(B)** Brother
(C) Son **(D)** Cousin

Q34. Farah is the mother of Nabil. Nabil is the father of Sara. Sara is the sister of Yusuf. Yusuf is male. How is Yusuf related to Farah?

(A) Grandson **(B)** Brother
(C) Nephew **(D)** Son

SECTION 6: DIRECTION SENSE & SPATIAL VECTORS

Q35. Ravi faces North. He makes a quarter-turn to his right. Which direction does he face now?

(A) West **(B)** South
(C) East **(D)** North

Q36. Mina faces East. She turns left, then turns left again. Which direction does she face?

(A) North **(B)** West
(C) East **(D)** South

Q37. A child walks 4 m North, then 3 m East. In which direction is the child from the start?

(A) North-West **(B)** South-West
(C) South-East **(D)** North-East

Q38. Omar faces West. He makes three quarter-turns to his right. Which direction does he face?

(A) West **(B)** North
(C) South **(D)** East

Q39. Arun faces North. He turns 180 degrees, then makes a quarter-turn left. Which direction does he face?

(A) West **(B)** North
(C) South **(D)** East

Q40. At the start, a toy faces East. It turns 270 degrees clockwise and then 180 degrees anticlockwise. Which direction does it face?

(A) West **(B)** South
(C) North **(D)** East

SECTION 7: TIME, CLOCKS & CALENDAR REASONING

Q41. A lesson starts at 9:35 a.m. and lasts 45 minutes. When does it end?

(A) 10:20 a.m. **(B)** 10:25 a.m.
(C) 10:10 a.m. **(D)** 10:15 a.m.

Q42. A bus must reach school at 8:15 a.m. The journey takes 50 minutes. At what time must it leave?

(A) 7:15 a.m. **(B)** 8:05 a.m.
(C) 7:35 a.m. **(D)** 7:25 a.m.

Q43. Sports practice starts at 4:10 p.m. and ends at 5:05 p.m. How many minutes does it last?

(A) 55 minutes **(B)** 45 minutes
(C) 50 minutes **(D)** 65 minutes

Q44. A clock is 8 minutes slow. When the correct time is 9:05 a.m., what does the clock show?

(A) 8:53 a.m. **(B)** 8:57 a.m.
(C) 9:05 a.m. **(D)** 9:13 a.m.

Q45. What is the smaller angle between the clock hands at exactly 6:30?

(A) 15 degrees **(B)** 180 degrees
(C) 30 degrees **(D)** 0 degrees

SECTION 8: QUANTITATIVE LOGIC & MATHEMATICAL PUZZLES

Q46. Six notebooks cost Rs 150. At the same price per notebook, what do 10 notebooks cost?

(A) Rs 300 **(B)** Rs 200
(C) Rs 250 **(D)** Rs 225

Q47. A pack of 5 identical pencils costs Rs 40. How many pencils can be bought for Rs 96 if pencils are also sold individually at the same rate?

(A) 10 **(B)** 12
(C) 11 **(D)** 15

Q48. A box costs Rs 18. Ribbon for each box costs another Rs 7. What is the total cost of preparing 6 boxes?

(A) Rs 168 **(B)** Rs 150
(C) Rs 108 **(D)** Rs 126

Q49. A shopkeeper buys a toy for Rs 240 and sells it for Rs 300. What is the profit percentage on the cost price?

(A) 30% **(B)** 20%
(C) 60% **(D)** 25%

Q50. At the same rate, 12 stickers cost as much as 3 badges. If one badge costs Rs 28, what is the price of one sticker?

(A) Rs 7 **(B)** Rs 9
(C) Rs 6 **(D)** Rs 8

SPACE FOR ROUGH WORK & SCRATCHPAD CALCULATIONS

ANSWER KEY & DETAILED STEP-BY-STEP SOLUTIONS

D3V3NDRA Classes • Class 7 Olympiad Logical Reasoning Master Solutions Guide

QUICK ANSWER KEY MATRIX (ALL 50 QUESTIONS)

1	2	3	4	5	6	7	8	9	10
D	**A**	**A**	**A**	**D**	**B**	**C**	**D**	**D**	**D**
11	12	13	14	15	16	17	18	19	20
C	**A**	**C**	**D**	**C**	**C**	**B**	**D**	**D**	**B**
21	22	23	24	25	26	27	28	29	30
C	**D**	**A**	**B**	**B**	**B**	**C**	**D**	**C**	**B**
31	32	33	34	35	36	37	38	39	40
A	**A**	**A**	**A**	**C**	**B**	**D**	**C**	**D**	**B**
41	42	43	44	45	46	47	48	49	50
A	**D**	**A**	**B**	**A**	**C**	**B**	**B**	**D**	**A**

COMPREHENSIVE STEP-BY-STEP LOGICAL DEDUCTIONS — PART 1 (QUESTIONS 01 TO 18)

Q1 [Q026] **Correct: (D) Option D**

Logical Step: The vertical mirror reverses left and right but keeps the top at the top. The character order becomes R2F, and each character is also reflected sideways. Merely typing the characters in reverse order does not produce a true mirror image. Compare both the order and the letter shapes with the answer diagram.

Q2 [Q027] **Correct: (A) Option A**

Logical Step: The vertical mirror reverses left and right but keeps the top at the top. The character order becomes DRIB, and each character is also reflected sideways. Merely typing the characters in reverse order does not produce a true mirror image. Compare both the order and the letter shapes with the answer diagram.

Q3 [Q031] **Correct: (A) Option A**

Logical Step: The vertical mirror reverses left and right but keeps the top at the top. The character order becomes G6B, and each character is also reflected sideways. Merely typing the characters in reverse order does not produce a true mirror image. Compare both the order and the letter shapes with the answer diagram.

Q4 [Q032] **Correct: (A) Option A**

Logical Step: The vertical mirror reverses left and right but keeps the top at the top. The character order becomes NEP, and each character is also reflected sideways. Merely typing the characters in reverse order does not produce a true mirror image. Compare both the order and the letter shapes with the answer diagram.

Q5 [Q036] **Correct: (D) Option D**

Logical Step: A vertical mirror swaps left and right, keeping each part at the same height. The black dot moves from top-right to top-left. The arrow pointing left must point right. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q6 [Q037] **Correct: (B) Option B**

Logical Step: A vertical mirror swaps left and right, keeping each part at the same height. The black dot moves from bottom-right to bottom-left. The arrow pointing down must point down. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q7 [Q038] **Correct: (C) Option C**

Logical Step: A vertical mirror swaps left and right, keeping each part at the same height. The black dot moves from top-left to top-right. The arrow pointing right must point left. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q8 [Q042] **Correct: (D) Option D**

Logical Step: A vertical mirror swaps left and right, keeping each part at the same height. The black dot moves from top-left to top-right. The arrow pointing right must point left. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q9 [Q051] **Correct: (D) Option D**

Logical Step: The horizontal water line reverses top and bottom but keeps left and right fixed. The characters stay in the left-to-right order F3L, while each character is flipped vertically. Compare the tops and bottoms of the letters with the answer diagram; a half-turn would also reverse their order and is incorrect.

Q10 [Q052] **Correct: (D) Option D**

Logical Step: The horizontal water line reverses top and bottom but keeps left and right fixed. The characters stay in the left-to-right order READ, while each character is flipped vertically. Compare the tops and bottoms of the letters with the answer diagram; a half-turn would also reverse their order and is incorrect.

Q11 [Q056] **Correct: (C) Option C**

Logical Step: The horizontal water line reverses top and bottom but keeps left and right fixed. The characters stay in the left-to-right order G2P, while each character is flipped vertically. Compare the tops and bottoms of the letters with the answer diagram; a half-turn would also reverse their order and is incorrect.

Q12 [Q057] **Correct: (A) Option A**

Logical Step: The horizontal water line reverses top and bottom but keeps left and right fixed. The characters stay in the left-to-right order LEAF, while each character is flipped vertically. Compare the tops and bottoms of the letters with the answer diagram; a half-turn would also reverse their order and is incorrect.

Q13 [Q061] **Correct: (C) Option C**

Logical Step: A water image swaps top and bottom, keeping each part in the same left-right position. The black dot moves from bottom-left to top-left. The arrow pointing down must point up. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q14 [Q062] **Correct: (D) Option D**

Logical Step: A water image swaps top and bottom, keeping each part in the same left-right position. The black dot moves from top-left to bottom-left. The arrow pointing right must point right. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q15 [Q063] **Correct: (C) Option C**

Logical Step: A water image swaps top and bottom, keeping each part in the same left-right position. The black dot moves from bottom-right to top-right. The arrow pointing up must point down. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q16 [Q067] **Correct: (C) Option C**

Logical Step: A water image swaps top and bottom, keeping each part in the same left-right position. The black dot moves from bottom-right to top-right. The arrow pointing up must point down. All other parts must undergo the same reflection; the matching diagram is shown alongside.

Q17 [Q076] **Correct: (B) 31**

Logical Step: The difference is always +5: 11 - 6 = 5 and 16 - 11 = 5. Continue by adding 5 to 26: 26 + 5 = 31.

Q18 [Q081] **Correct: (D) 36**

Logical Step: These are consecutive squares: 1 x 1, 2 x 2, 3 x 3, 4 x 4 and 5 x 5. The next is 6 x 6 = 36.

COMPREHENSIVE STEP-BY-STEP LOGICAL DEDUCTIONS — PART 2 (QUESTIONS 19 TO 34)

Q19 [Q085] **Correct: (D) 43**

Logical Step: The differences are 4, 6, 8 and 10. Increase the difference by 2 again to get 12. Then 31 + 12 = 43.

Q20 [Q088] **Correct: (B) 25**

Logical Step: Work out what is subtracted: 50 - 47 = 3, 47 - 43 = 4, 43 - 38 = 5 and 38 - 32 = 6. Next subtract 7, giving 32 - 7 = 25.

Q21 [Q092] **Correct: (C) 3**

Logical Step: Write the terms in halves: 1/2, 2/2, 3/2, 4/2 and 5/2. The numerator increases by 1 each time. The next term is 6/2, which simplifies to 3.

Q22 [Q096] **Correct: (D) 26**

Logical Step: Alternate two operations: multiply by 2, then subtract 2. After 14 x 2 = 28, the next operation is 28 - 2 = 26.

Q23 [Q176] **Correct: (A) EPH**

Logical Step: Move each letter forward one position: D becomes E, O becomes P, and G becomes H. Combining them gives EPH.

Q24 [Q177] **Correct: (B) ICOG**

Logical Step: Apply +2 to every letter: G to I, A to C, M to O, E to G. Thus GAME becomes ICOG.

Q25 [Q181] **Correct: (B) CAT**

Logical Step: Convert each number into a letter: 3 is C, 1 is A and 20 is T. Reading in order gives CAT.

Q26 [Q186] **Correct: (B) HPE**

Logical Step: First reverse DOG to get GOD. Next shift each letter forward one: G to H, O to P, D to E. This gives HPE.

Q27 [Q189] **Correct: (C) NHQM**

Logical Step: Use +2, -1, +2, -1 in order. L becomes N, I becomes H, O becomes Q, and N becomes M. Hence LION becomes NHQM.

Q28 [Q193] **Correct: (D) 41**

Logical Step: H=8, O=15, M=13 and E=5. The total is 8 + 15 + 13 + 5 = 41.

Q29 [Q001] **Correct: (C) Grandmother**

Logical Step: Asha is one generation above Rohan, and Rohan is one generation above Meera. Asha is therefore Meera's father's mother: her grandmother.

Q30 [Q005] **Correct: (B) Cousin**

Logical Step: Rita and Sita are sisters, so their children belong to two branches of the same family. Kabir is Rita's child and Anu is Sita's child. Children of siblings are cousins.

Q31 [Q008] **Correct: (A) Niece**

Logical Step: Kavya is Neel's sister. Isha is Kavya's daughter, so Isha is the daughter of Neel's sister. A sibling's daughter is called a niece; therefore Isha is Neel's niece.

Q32 [Q011] **Correct: (A) Aunt**

Logical Step: The woman and Sameer's mother have the same father. Since the woman is not his mother, she is his mother's sister and hence his aunt.

Q33 [Q015] **Correct: (A) Nephew**

Logical Step: Seema is Pooja's sister, and Anil is Seema's son. Thus Anil is the son of Pooja's sister. A sister's son is a nephew, so Anil is Pooja's nephew.

Q34 [Q023] **Correct: (A) Grandson**

Logical Step: Sara and Yusuf are siblings, so Nabil is also Yusuf's father. Farah is Nabil's mother. Thus Yusuf is Farah's male grandchild, or grandson.

COMPREHENSIVE STEP-BY-STEP LOGICAL DEDUCTIONS — PART 3 (QUESTIONS 35 TO 50)

Q35 [Q126] **Correct: (C) East**

Logical Step: A quarter-turn is 90 degrees. The clockwise compass order is North, East, South, West. Turning right means following this order one step, so a right turn from North ends at East.

Q36 [Q127] **Correct: (B) West**

Logical Step: Track one turn at a time. A left turn from East points North. A second left turn from North points West. Two quarter-turns have reversed the original facing direction.

Q37 [Q131] **Correct: (D) North-East**

Logical Step: The final position is both north and east of the start. Combining these two directions gives North-East.

Q38 [Q135] **Correct: (C) South**

Logical Step: Track each right turn: West to North, North to East, East to South. After three turns he faces South.

Q39 [Q139] **Correct: (D) East**

Logical Step: First complete the half-turn: 180 degrees changes North to South. Now use the new facing direction. A person facing South has East on their left, so the final direction is East.

Q40 [Q148] **Correct: (B) South**

Logical Step: A 270-degree clockwise turn is three right quarter-turns: East to South to West to North. A further 180-degree turn changes North to South.

Q41 [Q151] **Correct: (A) 10:20 a.m.**

Logical Step: From 9:35 to 10:00 is 25 minutes. There are 45 - 25 = 20 minutes left. Add these to 10:00 to get 10:20 a.m.

Q42 [Q154] **Correct: (D) 7:25 a.m.**

Logical Step: Work backward 50 minutes from 8:15. Subtract 15 minutes to reach 8:00, then the remaining 35 minutes to reach 7:25 a.m.

Q43 [Q157] **Correct: (A) 55 minutes**

Logical Step: Split the interval at the next hour. From 4:10 p.m. to 5:00 p.m. is 50 minutes. From 5:00 to 5:05 is another 5 minutes. Add them: 50 + 5 = 55 minutes.

Q44 [Q161] **Correct: (B) 8:57 a.m.**

Logical Step: A slow clock shows an earlier time. Subtract 8 minutes from 9:05: go back 5 minutes to 9:00, then 3 more minutes to 8:57 a.m.

Q45 [Q165] **Correct: (A) 15 degrees**

Logical Step: At 6:30, the minute hand is at 180 degrees from 12. The hour hand has moved halfway from 6 to 7: 180 + 15 = 195 degrees. Their smaller separation is 195 - 180 = 15 degrees.

Q46 [Q101] **Correct: (C) Rs 250**

Logical Step: One notebook costs 150 / 6 = Rs 25. Therefore 10 notebooks cost 10 x 25 = Rs 250.

Q47 [Q104] **Correct: (B) 12**

Logical Step: One pencil costs 40 / 5 = Rs 8. Divide the budget by the unit price: 96 / 8 = 12 pencils.

Q48 [Q107] **Correct: (B) Rs 150**

Logical Step: Each prepared box costs 18 + 7 = Rs 25. Six prepared boxes cost 6 x 25 = Rs 150.

Q49 [Q111] **Correct: (D) 25%**

Logical Step: Profit = 300 - 240 = Rs 60. Profit percentage is profit divided by cost price, times 100: 60 / 240 x 100 = 25%.

Q50 [Q115] **Correct: (A) Rs 7**

Logical Step: Three badges cost 3 x 28 = Rs 84, which is also the cost of 12 stickers. One sticker costs 84 / 12 = Rs 7.

CANDIDATE PERFORMANCE EVALUATION & BENCHMARK SCORECARD

1. Mirror (8 Qs)	2. Water (8 Qs)	3. Series (6 Qs)	4. Coding (6 Qs)	5. Blood (6 Qs)	6. Direction (6 Qs)	7. Clocks (5 Qs)	8. Quantitative (5 Qs)
____ / 16	____ / 16	____ / 12	____ / 12	____ / 12	____ / 12	____ / 10	____ / 10

Evaluator's Signature & Remarks | Total Score: _____ / 100 | Parent / Guardian Signature